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Subject: DSA

DSA Lab assignment 5

09/10

* Title: Write a C++ Program to demonstrate push, pop, and display operations performed on stack.

Q.1) What is stack data structure? Enlist and explain in detail the operations of stack.

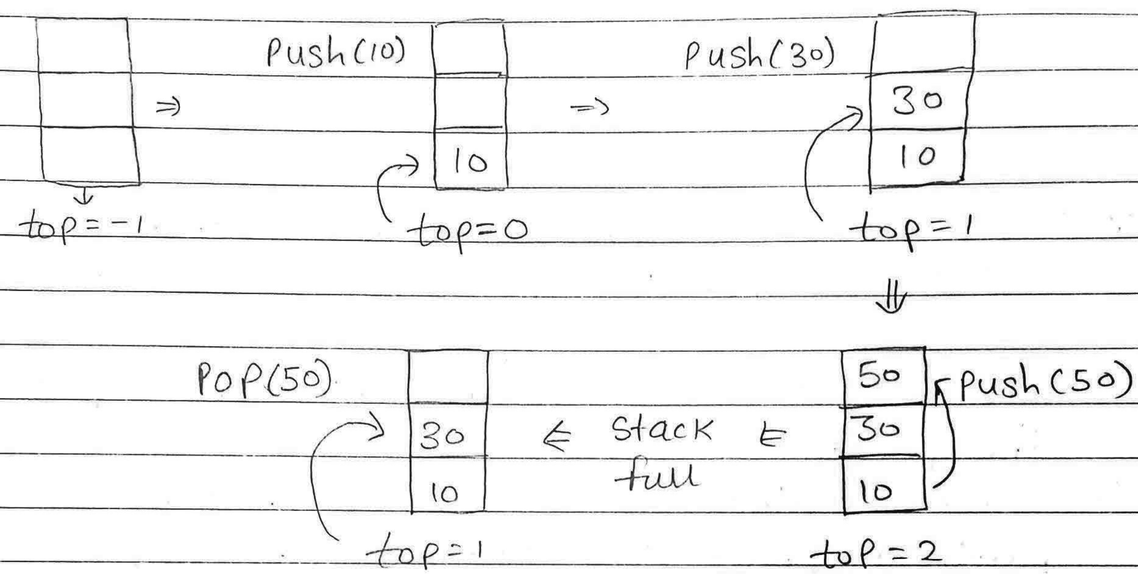
Sol^n:- Stack is a linear data structure that follows a particular order LIFO (Last in first out).

=) operations performed on a stack.

- Push():- adds an element to the top of the stack after inserting the element, increments top by 1.
- Pop():- removes the top element from the Stack and after removing the elements, decrements top by 1.
- Peak():- returns the top element without removing it from the Stack and displays the elements.
- Isempty():- check if the stack is empty by comparing top to -1 and returns true or false.
- Isfull():- checks if the stack is full (in case of fixed size arrays) by comparing ~~if~~ to size-1 and returns either true or false.

Q.2) Explain with neat diagram stack data structure?
Enlist and explain applications of stack data structure.

Solⁿ:- let array size = 3



⇒ Application of stack data structure:-

i) Recursion:- uses stack memory to store function calls and return memory addresses.

ii) Expression Evaluation:- stack is used to evaluate infix prefix and postfix expressions.

iii) Depth first search:- stack keep tracks of visited node in a graph used for backtracking, recursion.

iv) Browser History:- used for back and forward navigation in the browser using two different stacks.

v) function calls:- call stack returns local variables, parameters and return points.

vi) undo-Redo:- Action are stored in stack to reverse or reapply changes.

Q.3) Write an algorithm of push() and pop() operations on stack data structure.

Solⁿ:- push():

Step - (i) checks if stack is full.

Step - (ii) If stack is full, produces an error and exit.

Step - (iii) If stack is not full, increments top to point next empty space.

Step - (iv) Adds data element to the stack location where top is pointing.

(v) Return success.

Pop():

Step - (i) check if the stack is empty.

Step - (ii) If stack is empty, produces an error and exit

Step - (iii) If stack is not empty, accesses the data element at top

Step - (iv) Decreases value of top by 1.

Step - (v) Return Success.

Q.4) Write Pseudocode for push(), pop() and display() operations of stack data structures.

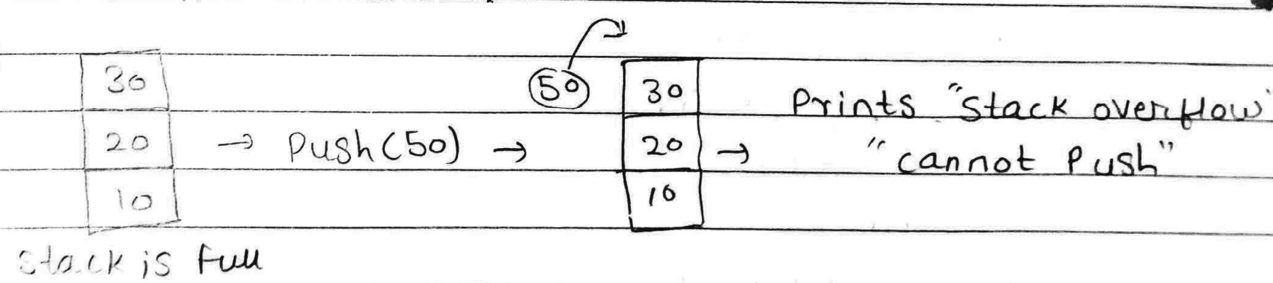
Solⁿ:-
```
function display():
    if isempty() is True:
        Print "stack is empty"
        Return
    Print "stack elements"
    for i from 0 to top:
        print arr[i]
```

```
function push(value):
    if isfull() is True:
        Print "Stack overflow, cannot push value".
    Else:
        Increment top by 1
        arr[top]= value
        Print value, "Pushed to the Stack.

function pop():
    if isempty() is True:
        Print "stack underflow, cannot Pop".
    Else:
        Print arr[top], "Popped from the stack".
        Decrement top by 1.
```

Q.5) What happen when you try to push an element into a full stack? Explain with an example

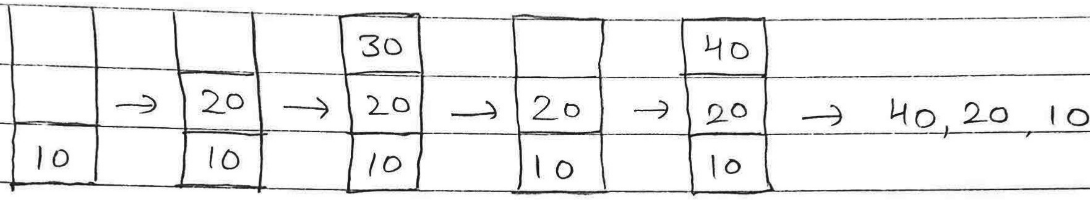
Sol^n:- when you try to push an element into a full stack, it will print "stack overflow" and will not push any value onto the stack.

=> 

=> Isfull() will return true and it won't be able to push value to the stack and will not process further.

Q.6) Simulate following sequence of stack operations
Push(10), Push(20), Push(30), Pop(), Push(40), display.

Sol^n:-



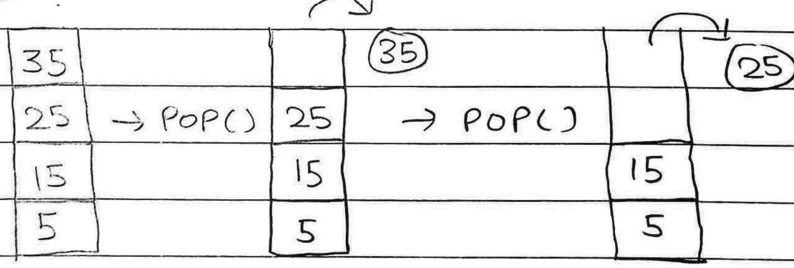
Q.7) What is stack underflow? How does program handle popping from an empty stack.

Sol^n:-

Stack underflow is when the stack is empty and pop() operation is called through function calling.
When isempty() is called and it returns true, program prints "stack underflow" and when stack is empty, pop() cannot be executed as there is no value so it exits the operation and prints "can't pop from empty stack".

Q.8) If stack currently contains elements [5,15,25,35] from bottom to top, what will be the results of two pop operations?

Sol^n:- Stack:



⇒ Stack now only has two elements [5,15].

Conclusion:- Thus, we have implemented stack operations using an array in C++.